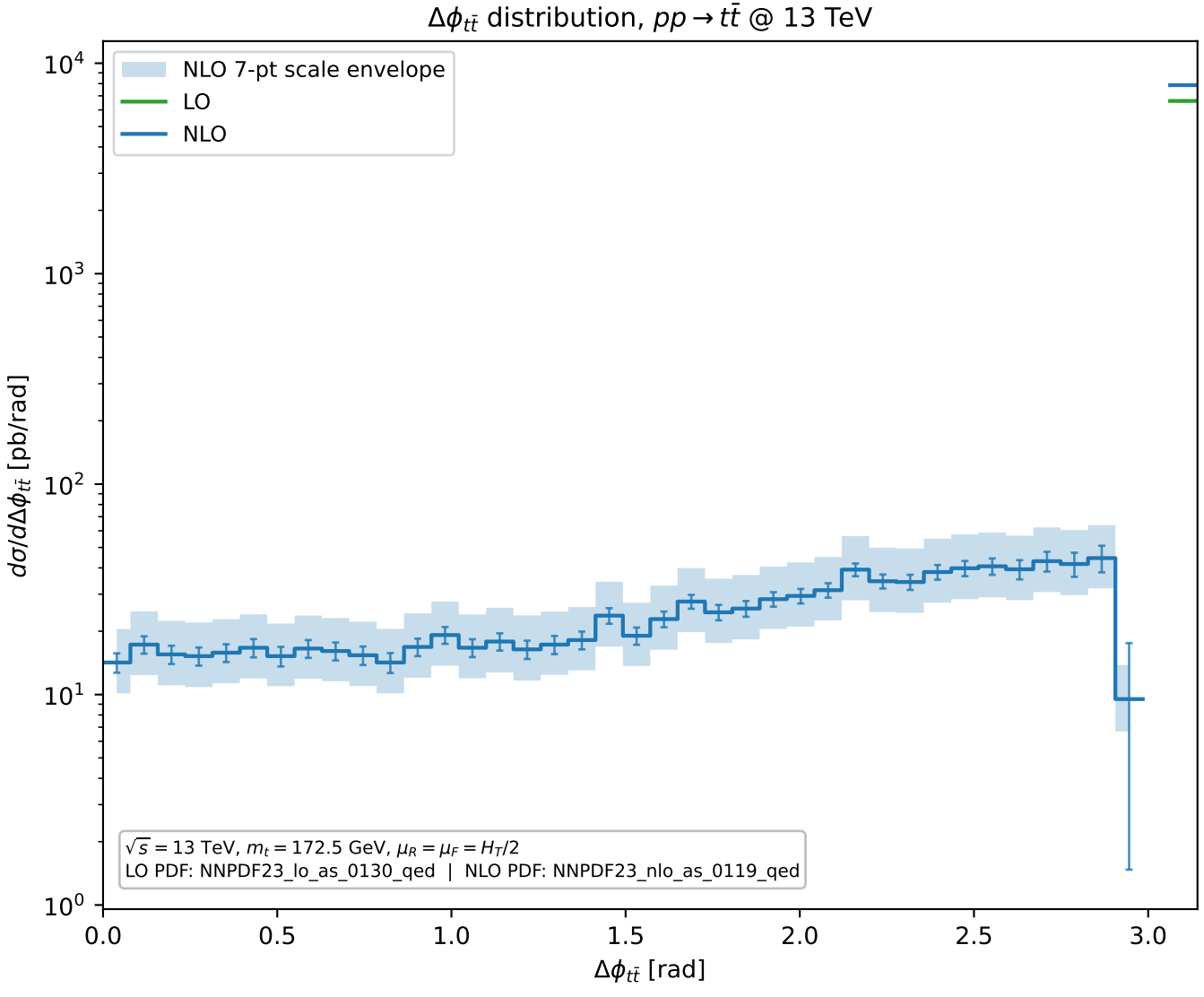

LO is an exact delta function at $\Delta\phi_{t\bar{t}} = \pi$ (all 100000/100000 LO events): last-bin entry only, $d\sigma/d\Delta\phi$=6618.96 pb/rad (=σ_{LO}/binwidth). NLO shows a genuine nonzero tail below pi from real emission. Ratio panel omitted (LO is a delta function). The second-to-last bin (2.985-3.063 rad, immediately below the pi-containing last bin) is negative and statistically consistent with zero within its (comparable-magnitude) error (-9.52+/-10.50 pb/rad, Neff=0.82) -- heavy positive/negative-weight cancellation immediately below the LO boundary; omitted from the log-scale line.